Лабораторная работа № 4
Первоначальное конфигурирование сети

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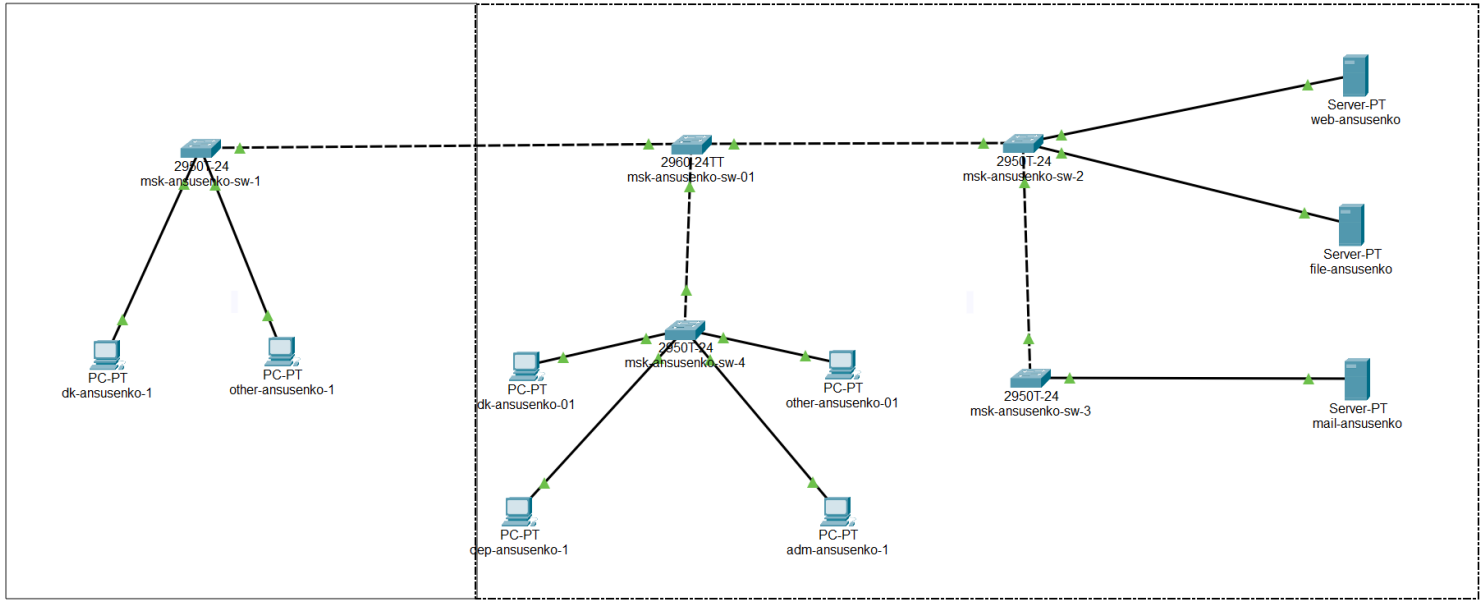
Цель работы

- Провести подготовительную работу по первоначальной настройке коммутаторов сети

Схема подключения оборудования



```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-ansusenko-sw-01
msk-ansusenko-sw-01(config)#interface vlan2
msk-ansusenko-sw-01(config-if)#no shutdown
msk-ansusenko-sw-01(config-if)#ip address 10.128.1.2 255.255.255.0
msk-ansusenko-sw-01(config-if)#exit
msk-ansusenko-sw-01(config)#ip default gateway 10.128.1.1
                                           ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-01(config)#ip default-gateway 10.128.1.1
msk-ansusenko-sw-01(config)#line vty 0 4
msk-ansusenko-sw-01(config-line)#password cisco
msk-ansusenko-sw-01(config-line)#login
msk-ansusenko-sw-01(config-line)#exit
msk-ansusenko-sw-01(config)#line console 0
msk-ansusenko-sw-01(config-line)#password cisco
msk-ansusenko-sw-01(config-line)#login
msk-ansusenko-sw-01(config-line)#exit
msk-ansusenko-sw-01(config)#enable secret cisco
msk-ansusenko-sw-01(config)#service password encryption
                                             ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-01(config)#service password-encryption
msk-ansusenko-sw-01(config)#username admin privilege 1 secret cisco
msk-ansusenko-sw-01(config)#ip domain-name ansusenko.rudn.edu
msk-ansusenko-sw-01(config)#crypto key generate rsa
The name for the keys will be: msk-ansusenko-sw-01.ansusenko.rudn.edu
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-ansusenko-sw-01(config)#line vty 0 4
*Mar 1 0:18:51.950: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:18:51.952: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-ansusenko-sw-01(config-line)#transport input ssh
msk-ansusenko-sw-01(config-line)#exit
msk-ansusenko-sw-01(config)#exit
msk-ansusenko-sw-01#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-ansusenko-sw-01#
```

```
msk-ansusenko-sw-01#show running-config
Building configuration...

Current configuration : 1580 bytes
!
version 12.2
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-ansusenko-sw-01
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 1
ip domain-name ansusenko.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport trunk allowed vlan 2
 switchport mode trunk
!
interface FastEthernet0/2
!
interface FastEthernet0/3
```

```
!
interface GigabitEthernet0/1
 switchport trunk allowed vlan 2
 switchport mode trunk
!
interface GigabitEthernet0/2
 switchport trunk allowed vlan 2
 switchport mode trunk
!
interface Vlan1
 no ip address
 shutdown
!
interface Vlan2
 ip address 10.128.1.2 255.255.255.0
!
ip default-gateway 10.128.1.1
!
!
!
!
line con 0
 password 7 0822455D0A16
 login
!
```

```
msk-ansusenko-sw-01#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down

msk-ansusenko-sw-01#show ssh
%No SSHv2 server connections running.
%No SSHv1 server connections running.
msk-ansusenko-sw-01#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24
2    Management                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
2    enet  100002     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0
```

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#enable
% Incomplete command.
Switch(config)#hostname msk-ansusenko-sw1
msk-ansusenko-sw1(config)#hostname msk-ansusenko-sw-1
msk-ansusenko-sw-1(config)#interface vlan2
msk-ansusenko-sw-1(config-if)#no shutdown
msk-ansusenko-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-ansusenko-sw-1(config-if)#exit
msk-ansusenko-sw-1(config)#ip default gateway 10.128.1.1
                                ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-1(config)#ip default-gateway 10.128.1.1
msk-ansusenko-sw-1(config)#line vty 0 4
msk-ansusenko-sw-1(config-line)#password cisco
msk-ansusenko-sw-1(config-line)#login
msk-ansusenko-sw-1(config-line)#exit
msk-ansusenko-sw-1(config)#line console 0
msk-ansusenko-sw-1(config-line)#password cisco
msk-ansusenko-sw-1(config-line)#login
msk-ansusenko-sw-1(config-line)#exit
msk-ansusenko-sw-1(config)#enable secret cisco
msk-ansusenko-sw-1(config)#service password encryption
                                            ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-1(config)#service password-encryption
msk-ansusenko-sw-1(config)#username admin privilege 1 secret cisco
msk-ansusenko-sw-1(config)#ip domain-name ansusenko.rudn.edu
msk-ansusenko-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-ansusenko-sw-1.ansusenko.rudn.edu
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-ansusenko-sw-1(config)#line vty 0 4
*Mar 1 0:23:28.341: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:23:28.341: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-ansusenko-sw-1(config-line)#transport input ssh
msk-ansusenko-sw-1(config-line)#exit
msk-ansusenko-sw-1(config)#exit
msk-ansusenko-sw-1#
```

```
msk-ansusenko-sw-1#show running-config
Building configuration...

Current configuration : 1567 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-ansusenko-sw-1
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 1
ip domain-name ansusenko.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!

msk-ansusenko-sw-1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual up                    up
FastEthernet0/21       unassigned      YES manual down                  down

msk-ansusenko-sw-1#show ssh
%No SSHv2 server connections running.
%No SSHv1 server connections running.
```

```
msk-ansusenko-sw-1#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/21, Fa0/22
                                                Fa0/23, Gig0/1, Gig0/2
2    Management                       active    Fa0/1, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
2    enet  100002     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0

msk-ansusenko-sw-1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      2

Port        Vlans allowed and active in management domain
Fa0/24      2

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      2
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-ansusenko-sw-2
msk-ansusenko-sw-2(config)#interface vlan2
msk-ansusenko-sw-2(config-if)#no shutdown
msk-ansusenko-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-ansusenko-sw-2(config-if)#exit
msk-ansusenko-sw-2(config)#ip default gateway 10.128.1.1
                                ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-2(config)#ip default-gateway 10.128.1.1
msk-ansusenko-sw-2(config)#line vty 0 4
msk-ansusenko-sw-2(config-line)#password cisco
msk-ansusenko-sw-2(config-line)#login
msk-ansusenko-sw-2(config-line)#transport input ssh
msk-ansusenko-sw-2(config-line)#exit
msk-ansusenko-sw-2(config)#line console 0
msk-ansusenko-sw-2(config-line)#password cisco
msk-ansusenko-sw-2(config-line)#login
msk-ansusenko-sw-2(config-line)#exit
msk-ansusenko-sw-2(config)#enable secret cisco
msk-ansusenko-sw-2(config)#service password-encryption
msk-ansusenko-sw-2(config)#username admin privilege 1 secret cisco
msk-ansusenko-sw-2(config)#ip domain-name ansusenko.rudn.edu
msk-ansusenko-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-ansusenko-sw-2.ansusenko.rudn.edu
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-ansusenko-sw-2(config)#ip ssh version 2
*Mar 1 1:11:21.803: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:11:21.803: %SSH-5-ENABLED: SSH 1.5 has been enabled
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
msk-ansusenko-sw-2(config)#line vty 0 4
msk-ansusenko-sw-2(config-line)#transport input ssh
msk-ansusenko-sw-2(config-line)#exit
msk-ansusenko-sw-2(config)#exit
msk-ansusenko-sw-2#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
msk-ansusenko-sw-2#
```

```
msk-ansusenko-sw-2#show running-config
Building configuration...

Current configuration : 1411 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-ansusenko-sw-2
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 1
ip domain-name ansusenko.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!

msk-ansusenko-sw-2#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual up                    up
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down

msk-ansusenko-sw-2#show ssh
%No SSHv2 server connections running.
```

```
msk-ansusenko-sw-2#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

msk-ansusenko-sw-2#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      auto         n-802.1q       trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1

msk-ansusenko-sw-2#
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname msk-ansusenko-sw-3
msk-ansusenko-sw-3(config)#interface vlan2
msk-ansusenko-sw-3(config-if)#no shutdown
msk-ansusenko-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-ansusenko-sw-3(config-if)#exit
msk-ansusenko-sw-3(config)#ip default gateway 10.128.1.1
                                          ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-3(config)#ip default-gateway 10.128.1.1
msk-ansusenko-sw-3(config)#line vty 0 4
msk-ansusenko-sw-3(config-line)#password cisco
msk-ansusenko-sw-3(config-line)#login
msk-ansusenko-sw-3(config-line)#transport input ssh
msk-ansusenko-sw-3(config-line)#exit
msk-ansusenko-sw-3(config)#transport input ssh
                           ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-3(config)#line console 0
msk-ansusenko-sw-3(config-line)#password cisco
msk-ansusenko-sw-3(config-line)#login
msk-ansusenko-sw-3(config-line)#exit
msk-ansusenko-sw-3(config)#enable secret cisco
msk-ansusenko-sw-3(config)#service password encryption
                                            ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-3(config)#service password-encryption
msk-ansusenko-sw-3(config)#username admin privilege 1 secret cisco
msk-ansusenko-sw-3(config)#ip domain-name ansusenko.rudn.edu
msk-ansusenko-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-ansusenko-sw-3.ansusenko.rudn.edu
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-ansusenko-sw-3(config)#line vty 0 4
*Mar 1 1:17:17.915: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 1:17:17.915: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-ansusenko-sw-3(config-line)#transport input ssh
msk-ansusenko-sw-3(config-line)#exit
msk-ansusenko-sw-3(config)#exit
msk-ansusenko-sw-3#
```

```
msk-ansusenko-sw-3#show running-config
Building configuration...

Current configuration : 1411 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-ansusenko-sw-3
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 1
ip domain-name ansusenko.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!

msk-ansusenko-sw-3#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual down                  down
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual down                  down
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual down                  down
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
```

```
msk-ansusenko-sw-3#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
```

```
Switch>enable
Switch#
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hostname msk-ansusenko-sw-4
msk-ansusenko-sw-4(config)#interface vlan2
msk-ansusenko-sw-4(config-if)#no shutdown
msk-ansusenko-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-ansusenko-sw-4(config-if)#exit
msk-ansusenko-sw-4(config)#ip default gateway 10.128.1.1
                                          ^
% Invalid input detected at '^' marker.

msk-ansusenko-sw-4(config)#ip default-gateway 10.128.1.1
msk-ansusenko-sw-4(config)#enable secret cisco
msk-ansusenko-sw-4(config)#service password-encryption
msk-ansusenko-sw-4(config)#username admin privilege 15 secret cisco
msk-ansusenko-sw-4(config)#ip domain-name ansusenko.rudn.edu
msk-ansusenko-sw-4(config)#crypto key generate rsa
The name for the keys will be: msk-ansusenko-sw-4.ansusenko.rudn.edu
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 512
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-ansusenko-sw-4(config)#ip ssh version 2
*Mar 1 0:29:41.3: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:29:41.3: %SSH-5-ENABLED: SSH 1.5 has been enabled
Please create RSA keys (of at least 768 bits size) to enable SSH v2.
msk-ansusenko-sw-4(config)#line console 0
msk-ansusenko-sw-4(config-line)#password cisco
msk-ansusenko-sw-4(config-line)#login
msk-ansusenko-sw-4(config-line)#logging synchronous
msk-ansusenko-sw-4(config-line)#exit
msk-ansusenko-sw-4(config)#line vty 0 4
msk-ansusenko-sw-4(config-line)#login local
msk-ansusenko-sw-4(config-line)#transport input ssh
msk-ansusenko-sw-4(config-line)#exit
msk-ansusenko-sw-4(config)#end
msk-ansusenko-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-ansusenko-sw-4#write memory
Building configuration...
[OK]
```

```
msk-ansusenko-sw-4#show running-config
Building configuration...

Current configuration : 1411 bytes
!
version 12.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname msk-ansusenko-sw-4
!
enable secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!
ip ssh version 1
ip domain-name ansusenko.rudn.edu
!
username admin secret 5 $1$mERr$hx5rVt7rPNoS4wqbXKX7m0
!
!
!

msk-ansusenko-sw-4#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual up                    up
FastEthernet0/2        unassigned      YES manual down                  down
FastEthernet0/3        unassigned      YES manual down                  down
FastEthernet0/4        unassigned      YES manual down                  down
FastEthernet0/5        unassigned      YES manual down                  down
FastEthernet0/6        unassigned      YES manual up                    up
FastEthernet0/7        unassigned      YES manual down                  down
FastEthernet0/8        unassigned      YES manual down                  down
FastEthernet0/9        unassigned      YES manual down                  down
FastEthernet0/10       unassigned      YES manual down                  down
FastEthernet0/11       unassigned      YES manual up                    up
FastEthernet0/12       unassigned      YES manual down                  down
FastEthernet0/13       unassigned      YES manual down                  down
FastEthernet0/14       unassigned      YES manual down                  down
FastEthernet0/15       unassigned      YES manual down                  down
FastEthernet0/16       unassigned      YES manual up                    up
FastEthernet0/17       unassigned      YES manual down                  down
FastEthernet0/18       unassigned      YES manual down                  down
FastEthernet0/19       unassigned      YES manual down                  down
FastEthernet0/20       unassigned      YES manual down                  down
FastEthernet0/21       unassigned      YES manual down                  down
FastEthernet0/22       unassigned      YES manual down                  down
FastEthernet0/23       unassigned      YES manual down                  down
FastEthernet0/24       unassigned      YES manual down                  down
```

```
msk-ansusenko-sw-4#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
1    enet  100001     1500  -      -      -        -    -        0      0
1002 fddi  101002     1500  -      -      -        -    -        0      0
1003 tr    101003     1500  -      -      -        -    -        0      0
1004 fdnet 101004     1500  -      -      -        ieee -        0      0
1005 trnet 101005     1500  -      -      -        ibm  -        0      0

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------

Remote SPAN VLANs
------------------------------------------------------------------------------

Primary Secondary Type              Ports
------- --------- ----------------- ------------------------------------------
msk-ansusenko-sw-4#
msk-ansusenko-sw-4#show interfaces trunk
```

Сводная информация по настройкам

В процессе выполнения работы все коммутаторы были настроены в соответствии с планом IP-адресации:

Устройство	**Имя**	**IP-адрес**
Коммутатор 1	msk-ansusenko-sw-01	10.128.1.2
Коммутатор 2	msk-ansusenko-sw-1	10.128.1.6
Коммутатор 3	msk-ansusenko-sw-2	10.128.1.3
Коммутатор 4	msk-ansusenko-sw-3	10.128.1.4
Коммутатор 4	msk-ansusenko-sw-4	10.128.1.5

Контрольные вопросы

Вопрос 1: При помощи каких команд можно посмотреть конфигурацию сетевого оборудования?

Для просмотра конфигурации сетевого оборудования Cisco используются следующие команды:

- `show running-config` (или `show run`) — отображает текущую конфигурацию
- `show startup-config` (или `show start`) — отображает стартовую конфигурацию
- `show ip interface brief` — показывает информацию по интерфейсам
- `show vlan` — отображает информацию о VLAN
- `show interfaces trunk` — показывает информацию о транковых портах

Контрольные вопросы

Вопрос 2: При помощи каких команд можно посмотреть стартовый конфигурационный файл оборудования?

Стартовый конфигурационный файл (startup-config) просматривается командой:

- `show startup-config`
- `show start` (сокращенный вариант)

Контрольные вопросы

Вопрос 3: При помощи каких команд можно экспортировать конфигурационный файл оборудования?

Для экспорта конфигурации используются следующие команды:

- `write memory`
- `copy running-config startup-config`
- `copy running-config tftp:`
- `copy startup-config tftp:`

Контрольные вопросы

Вопрос 4: При помощи каких команд можно импортировать конфигурационный файл оборудования?

Для импорта конфигурации используются следующие команды:

- `copy startup-config running-config`
- `copy tftp: running-config`
- `copy tftp: startup-config`
- `reload`

Выводы

- В результате выполнения работы мы успешно воссоздали схему подключения оборудования, настроили её и провели проверку текущих настроек.